Affordable Trash-Collecting and Sorting Amphibious Boat

Design report · October 2026

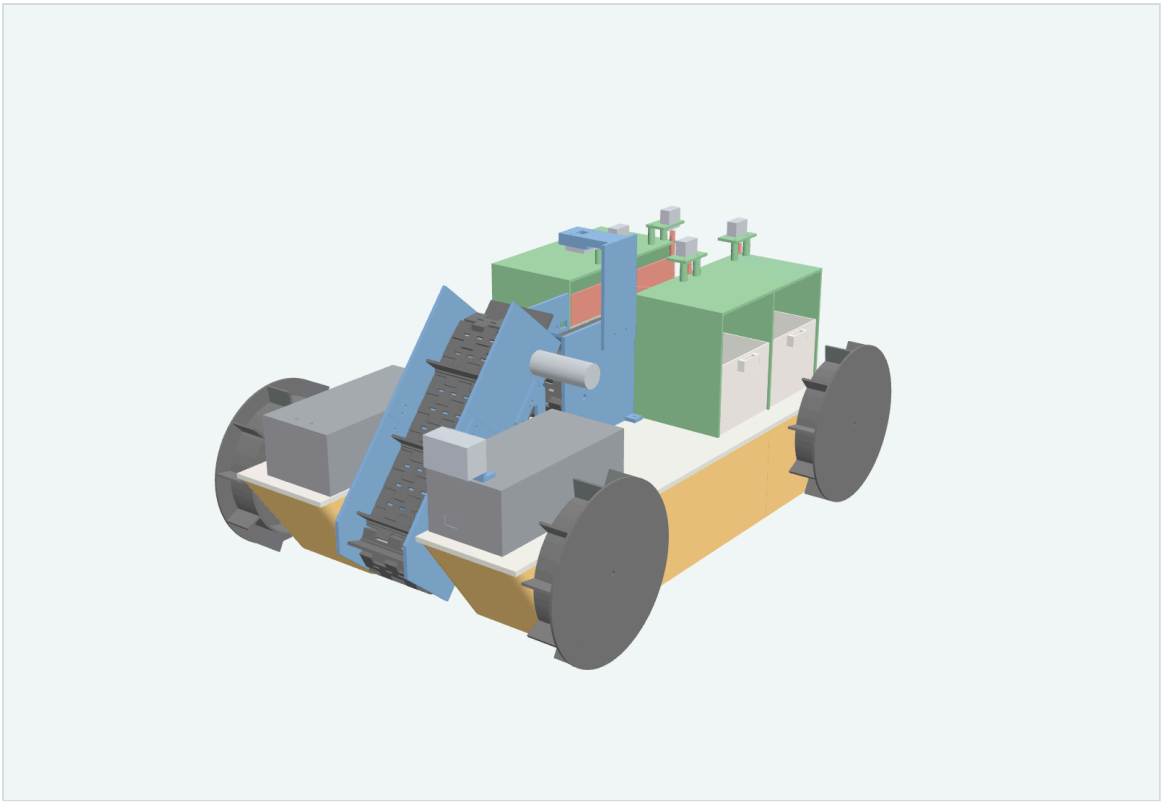


Figure 1. Assembled boat: bow intake conveyor, electronics boxes, sorting conveyor and bins.

1. Summary

This report describes an affordable, 3D-printed amphibious boat. A conveyor at the bow collects floating trash. A camera on a second conveyor identifies each item, and a servo-driven door sorts it into one of 4 bins. It drives on land and in water on 4 paddle wheels, and it is steered from a phone with a live navigation camera.

The design was modelled parametrically and checked in software: no parts overlap, every part fits a 220 × 220 mm printer bed, and the buoyancy was computed from the hull shape. Estimated materials cost is ≈ **₱12,960**, and **4.1 kg** of filament prints **127 parts**.

2. Design requirements

I still couldn't find a published OCNR rulebook. The design follows the brief you gave earlier, plus this request:

- 3D-printed boat that runs on land and in water.
- Front conveyor that collects trash.
- Camera on a second, top conveyor that sorts into up to 4 containers.

- Only 4 servos, all used to open and close the containers.
- A front-facing navigation camera.
- Parts available from MakerLab Electronics (makerlab.ph) or other local suppliers.

Check the size, weight and power limits against the official OCNR sheet.

3. How it works

1. **Intake conveyor (C1).** The conveyor sits between the two pontoons at a 50° angle. Its lower drum sits just below the waterline. Cleats every 4th link scoop floating items up out of the water, and holes in every link let the water drain.
2. **Transfer chute.** At the top, items slide down a short chute onto the sorting conveyor.
3. **Sorting conveyor (C2).** This conveyor runs flat toward the stern. An IR sensor stops both conveyors when an item reaches the camera zone. The Pi camera looks down from 108 mm above the belt, with an LED strip for even light, and classifies the item on 5 frames. The majority vote picks the bin.
4. **Sorting into bins.** Each bin is a box beside the belt with a door (flap) in its side wall. The door's servo swings it 60° across the belt. The moving belt pushes the item along the open door and into the bin, then the servo closes the door again. If the camera isn't sure (below 70% confidence), the item goes to bin 4 ("Other"). Anything that gets past every door falls into the reject tray at the stern.
5. **Steering.** You drive from a phone browser on the Pi's Wi-Fi. The page shows the navigation camera live, has a joystick and per-bin counters, and lets you open any bin by hand and unjam the intake. The drive motors stop if the phone link drops for 0.8 s.

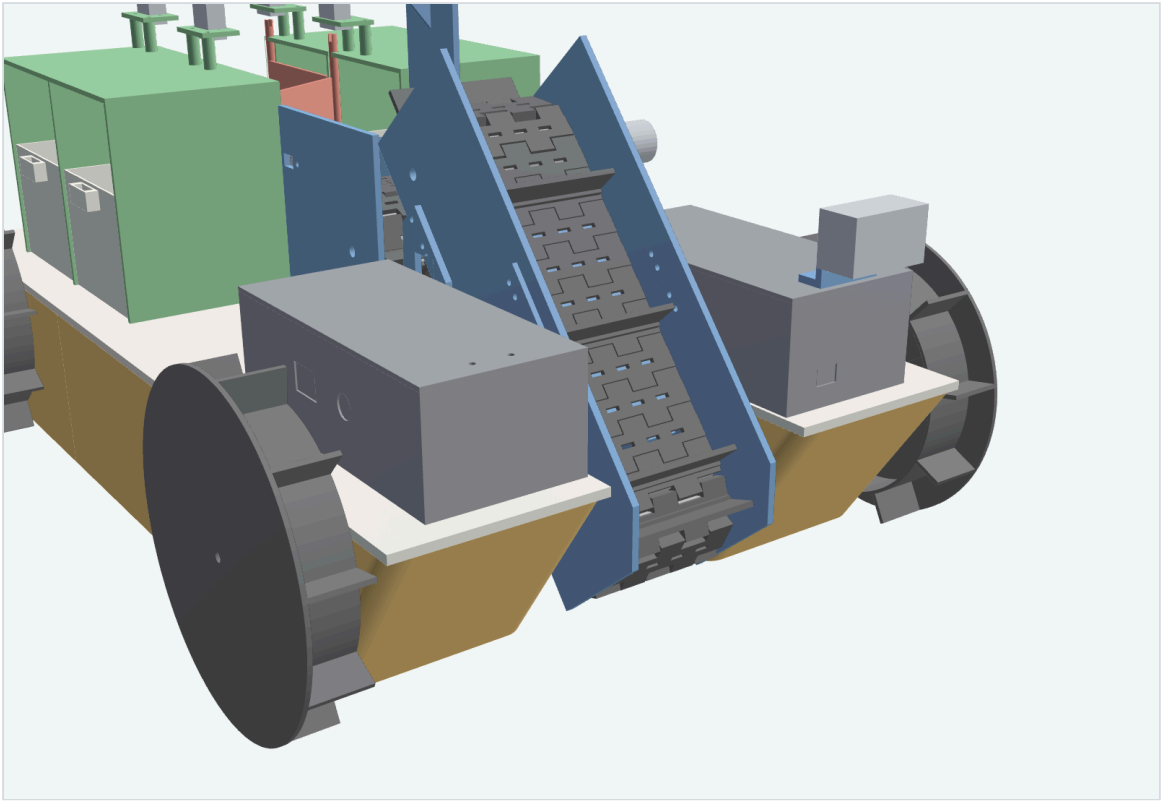


Figure 2. Cleated intake conveyor between the pontoons.

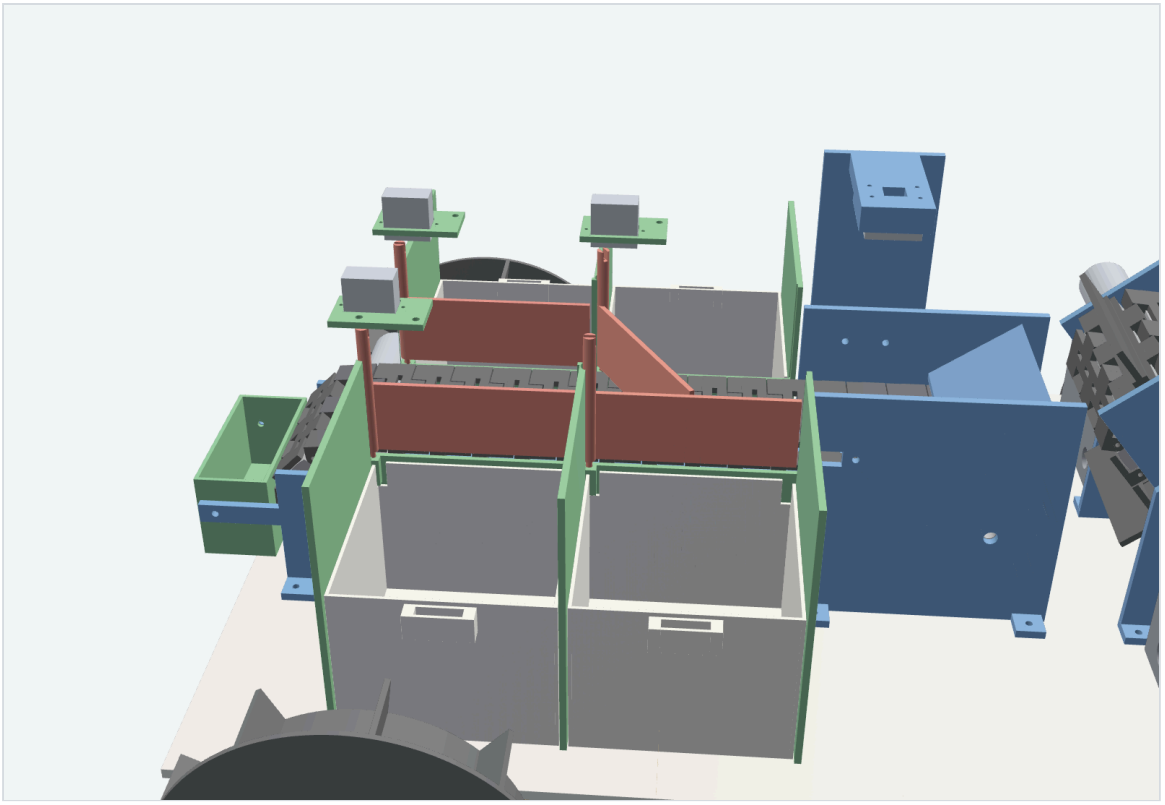


Figure 3. Sorting conveyor with one bin door open (covers hidden).

4. Specifications

Item	Value
Hull	Catamaran, 2 pontoons of 630 × 115 × 100 mm, each printed as 3 sealed segments
Overall size	650 L × 424 W × 315 H mm (on land)
Dry mass	≈ 6.0 kg (≈ 4.1 kg printed parts, 1.9 kg bought parts)
Buoyancy	13.1 kg to the deck, about 2× the boat's weight
Draft (how deep it sits)	47 mm dry, 51 mm with 500 g of trash. The drive motors sit at 65 mm, so they stay 14 mm above the water
Stability	Metacentric height ≈ 225 mm (very stable, because the pontoons are far apart)
Drive	4 × JGA25-370 12 V 60 rpm motors with Ø176 mm paddle wheels, tank steering. Ground clearance 10 mm
Intake belt C1	80 mm wide printed link belt, 29 links (7 with cleats), 50° incline, ≈ 64 mm/s
Sorting belt C2	80 mm wide, 39 links, 310 mm between drums, ≈ 56 mm/s
Bins	4 removable buckets, each 93 × 99 × 74 mm inside (≈ 0.65 L)
Item size	Up to about 75 mm wide, 70 mm long and 50 mm tall: caps, small bottles, cans, cups, wrappers, foam
Electronics	Raspberry Pi Zero 2 W, Pi camera (sorting), USB webcam (navigation), PCA9685 servo board, 2 × L298N, 3S1P 18650 battery
Run time	About 1 h on a 3S1P pack, at roughly 2 A average (about 2 h with a 3S2P pack)

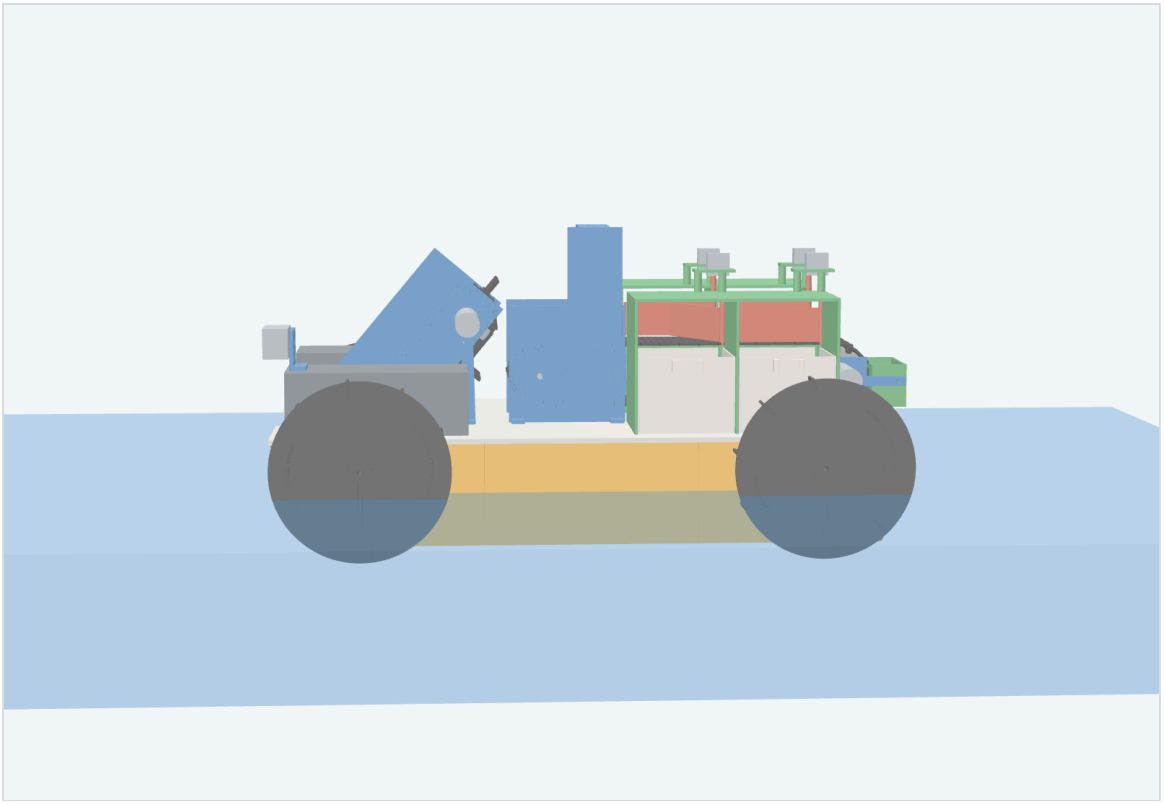


Figure 4. Side view at the computed waterline.

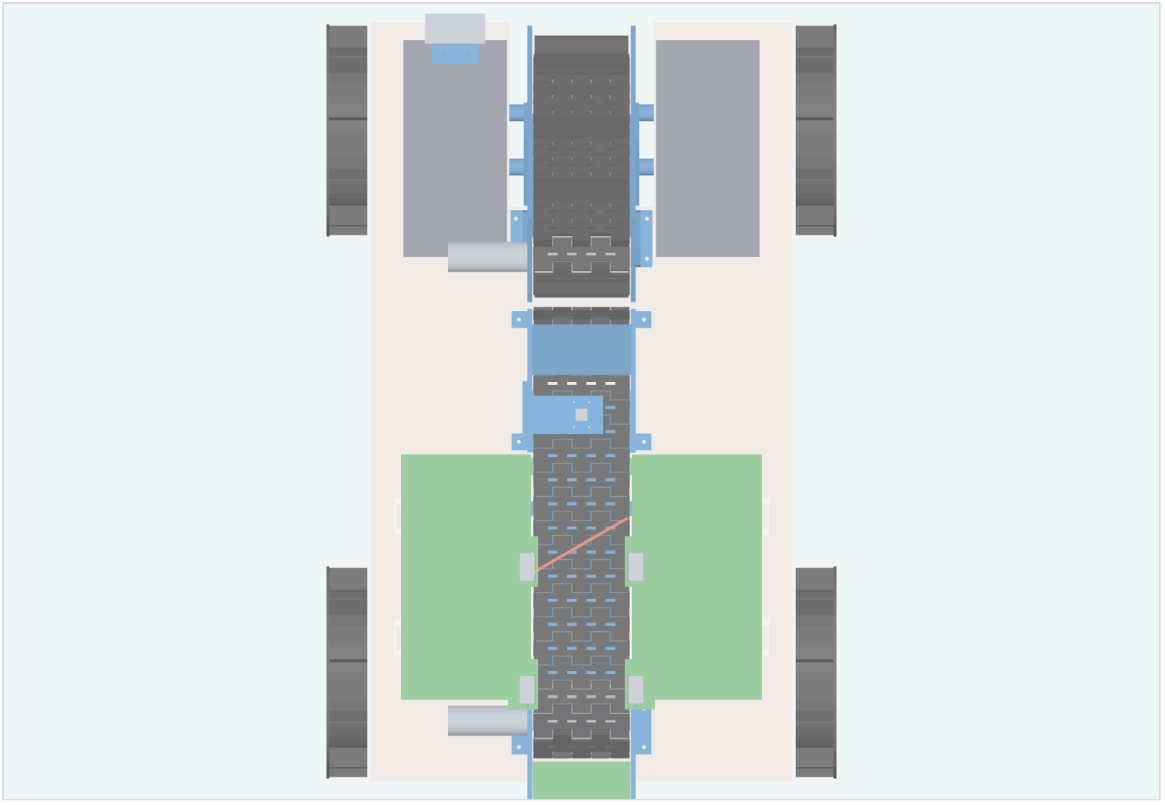


Figure 5. Plan view.

5. Engineering calculations

Buoyancy and draft

The boat sinks until the water it pushes aside weighs as much as the boat ($m = \rho \cdot V$). The waterline was found from the CAD pontoon volume by bisection.

Load	Mass (g)	Draft (mm)
Dry	6003	46.9
With 500 g of trash	6503	50.7
Maximum (water at the deck)	13098	100

Reserve buoyancy is 101% at full load. The drive-motor pockets start at z = 65 mm, which stays above the waterline.

Stability

$GM = KB + BM - KG$, with $BM = I_T / V$.

For the two pontoons (waterline length ≈ 600 mm, width 115 mm, pontoon centres at ±117.5 mm): $I_T \approx 2.06e{+}09$ mm⁴, so BM ≈ 343 mm. With KB ≈ 23 mm and an estimated KG ≈ 140 mm, **GM ≈ 226 mm**. That is a very stable boat: the wide pontoon spacing dominates.

Belt speed and sorting geometry

- **Belt speed:** each sprocket turn moves the belt 8 teeth × 20 mm = 160 mm. At 30 rpm that is 80 mm/s, so about 64 mm/s (intake) and 56 mm/s (sorting) at the PWM settings used.
- **Belts:** 68 printed links in total (intake 29, sorting 39). The sprocket pitch radius is 26.13 mm.
- **Doors:** each door is 95.8 mm long and swings 60° across the 80 mm belt. It stops about 1 mm short of the far edge and about 3 mm short of the opposite closed door, checked in the CAD assembly.

Power and run time

- Wheel motors: about 4 × 0.25 A.
- Conveyors: about 2 × 0.2 A.
- Pi Zero 2 W, both cameras and the LED strip: about 0.4 A at 12 V.
- Servos, idle: about 0.05 A.

That totals about **1.9 A** average. A 3S1P pack of 2.5 Ah cells gives **about 1 h** of run time, allowing for losses (about 2 h with a 3S2P pack).

6. Materials list and suppliers

Prices are approximate Philippine online prices (October 2026). Check them with the store before you buy.

Suppliers:

- **MakerLab Electronics** (makerlab.ph, also on Shopee and Lazada): Raspberry Pi, cameras, motors, drivers, servos, filament.
- **Circuitrocks** (circuit.rocks): Raspberry Pi and cameras.
- **e-Gizmo Mechatronix Central:** motor drivers, sensors, batteries.
- **Hardware stores** (Ace Hardware, Wilcon Depot, Handyman): threaded rod, screws, epoxy.
- **Sign shops:** Sintra (PVC foam) board.

Electronics

Part	Qty	Approx. ₱	Where
Raspberry Pi Zero 2 W (+ header pins)	1	1,700	MakerLab, Circuitrocks
microSD card 16 GB (Raspberry Pi OS 64-bit)	1	250	any
Pi camera OV5647 5 MP + Pi Zero camera cable (sorting camera)	1	500	MakerLab, Circuitrocks
USB webcam 480p/720p + micro-USB OTG adapter (navigation camera)	1	400	any computer shop
PCA9685 16-channel servo driver	1	250	MakerLab, e-Gizmo
SG90 micro servo (bin doors)	4	320	MakerLab
L298N dual motor driver (1 for drive, 1 for conveyors)	2	300	MakerLab, e-Gizmo
JGA25-370 gear motor, 12 V, 60 rpm (wheels)	4	1,800	MakerLab, Shopee
JGA25-370 gear motor, 12 V, 30 rpm (conveyors)	2	900	MakerLab, Shopee
LM2596 buck converter (one set to 5.1 V for the Pi, one to 5.5 V for servos + LEDs)	2	200	MakerLab
FC-51 IR obstacle sensor (item at camera)	1	50	MakerLab, e-Gizmo
5 V white LED strip, 20 cm (camera lighting)	1	80	any
18650 Li-ion cells, genuine 2,500 mAh (3S1P)	3	750	e-Gizmo, battery shops
3S 20 A BMS + nickel strip	1	150	e-Gizmo, Shopee
12.6 V 2 A Li-ion charger + DC jack	1	350	Shopee
Rocker switch, inline blade fuse holder + 10 A fuse	1	120	any
Hook-up wire, Dupont jumpers, JST plugs, heat shrink	1 set	300	any
Electronics subtotal		**≈ 8,420**	

Mechanical

Part	Qty	Approx. ₱	Where
PLA filament, 1.75 mm (4.1 kg used)	5 kg	3,250	MakerLab
Sintra (PVC foam) board, 5 mm, piece at least 650 × 370 mm (deck)	1	400	sign shop, Lazada
M6 threaded rod, 1 m (4 drum axles) + 12 M6 nuts and washers	1	150	hardware store

Part	Qty	Approx. ₱	Where
M3 screw and nut assortment (8–20 mm)	1	250	hardware store, MakerLab
M2 screws (servo tabs, horns) + 6 × M3 grub screws	1	110	MakerLab
Two-part epoxy (segment joints + a waterproof coat over the PLA)	1	250	hardware store
Hot glue sticks, electrical tape, petroleum jelly	1	130	any
Mechanical subtotal		**≈ 4,540**	
Total		**≈ ₱12,960**	

That is about ₱6,700 less than the first version of this design. The belt pins are 84 mm pieces of the same filament, so they cost nothing extra. The software is unchanged on the Pi Zero 2 W. It classifies about 2–3 frames per second, so each item takes about 2 s at the camera.

Optional upgrades:

- Raspberry Pi 4 (+₱4,300): faster sorting. Its board holes differ: change `PI_HOLES` in `cad/electronics.py` to 58 × 49 mm.
- MG90S metal-gear servos (+₱400).
- PETG instead of PLA (+₱1,000): handles water better.
- A 3S2P pack (+₱750): doubles the run time.

7. Printed parts

Files are in `02_Print_Files/`, one folder per subsystem. Each file name ends with the number of copies to print (`__x2`). `Print_List.csv` has the full list. No part needs supports.

Folder	Copies	Filament (g)	Settings
`1_Hull`	18	1480	PLA/PETG, 3 walls, 5 top + 5 bottom, 5% gyroid, no supports
`2_Drive_Wheels`	4	411	3 walls, 20% infill, flange down
`3_Intake_Conveyor`	37	552	2-3 walls, 15% infill; belt links on edge as exported
`4_Sorting_Conveyor`	46	628	2-3 walls, 15% infill; belt links on edge as exported
`5_Sorter_Bins`	16	731	2-3 walls, 15% infill
`6_Electronics_Mounts`	6	309	2-3 walls, 15% infill

Subsystem	File	Qty	Size (mm)	g	Note
Hull	`01_pontoon_bow_port__x1.stl`	1	210 x 115 x 100	222	
Hull	`02_pontoon_bow_stbd__x1.stl`	1	210 x 115 x 100	222	
Hull	`03_pontoon_mid__x2.stl`	2	210 x 115 x 100	516	one per side
Hull	`04_pontoon_stern_port__x1.stl`	1	210 x 115 x 100	245	

Subsystem	File	Qty	Size (mm)	g	Note
Hull	`05_pontoon_stern_stbd__x1.stl`	1	210 x 115 x 100	245	
Hull	`06_dowel__x8.stl`	8	6 x 6 x 22	4	segment joints
Hull	`07_motor_saddle__x4.stl`	4	26 x 40 x 22	26	clamps the drive motors
Drive Wheels	`01_paddle_wheel__x4.stl`	4	176 x 176 x 34	411	
Intake Conveyor	`01_c1_frame_lower__x1.stl`	1	120 x 96 x 137	110	
Intake Conveyor	`02_c1_frame_upper__x1.stl`	1	90 x 91 x 137	106	
Intake Conveyor	`03_c1_splice__x2.stl`	2	46 x 73 x 3	16	
Intake Conveyor	`04_c1_leg_port__x1.stl`	1	47 x 91 x 14	9	
Intake Conveyor	`05_c1_leg_stbd__x1.stl`	1	47 x 91 x 14	9	
Intake Conveyor	`06_drum_drive__x1.stl`	1	54 x 54 x 81	48	top drum
Intake Conveyor	`07_drum_idler__x1.stl`	1	54 x 54 x 81	48	bottom drum
Intake Conveyor	`08_belt_link__x22.stl`	22	27 x 10 x 80	136	
Intake Conveyor	`09_belt_link_cleat__x7.stl`	7	27 x 30 x 80	70	every 4th link
Sorting Conveyor	`01_c2_front_frame__x1.stl`	1	116 x 120 x 120	117	camera zone + chute
Sorting Conveyor	`02_c2_bed_beam__x1.stl`	1	192 x 84 x 41	127	
Sorting Conveyor	`03_c2_head_stand_drive__x1.stl`	1	84 x 65 x 17	10	port
Sorting Conveyor	`04_c2_head_stand_idler__x1.stl`	1	84 x 65 x 17	10	starboard
Sorting Conveyor	`05_drum_drive__x1.stl`	1	54 x 54 x 81	48	stern drum
Sorting Conveyor	`06_drum_idler__x1.stl`	1	54 x 54 x 81	48	front drum
Sorting Conveyor	`07_belt_link__x39.stl`	39	27 x 10 x 80	242	
Sorting Conveyor	`08_reject_tray__x1.stl`	1	36 x 81 x 44	26	
Sorter Bins	`01_sorter_frame_port__x1.stl`	1	204 x 120 x 131	133	bins 1 + 3
Sorter Bins	`02_sorter_frame_stbd__x1.stl`	1	204 x 120 x 131	133	bins 2 + 4
Sorter Bins	`03_sorter_cover_port__x1.stl`	1	204 x 108 x 23	54	
Sorter Bins	`04_sorter_cover_stbd__x1.stl`	1	204 x 108 x 23	54	
Sorter Bins	`05_servo_bridge__x4.stl`	4	42 x 25 x 3	9	
Sorter Bins	`06_flap_port__x2.stl`	2	98 x 72 x 6	18	bins 1 + 3
Sorter Bins	`07_flap_stbd__x2.stl`	2	98 x 72 x 6	18	bins 2 + 4
Sorter Bins	`08_bucket__x4.stl`	4	96 x 110 x 76	312	
Electronics Mounts	`01_electronics_box__x1.stl`	1	180 x 86 x 60	100	port bow deck
Electronics Mounts	`02_battery_box__x1.stl`	1	180 x 86 x 60	98	starboard bow deck
Electronics Mounts	`03_box_lid__x2.stl`	2	180 x 86 x 8	79	

Subsystem	File	Qty	Size (mm)	g	Note
Electronics Mounts	`04_camera_mast__x1.stl`	1	56 x 110 x 67	26	sorting camera + LEDs
Electronics Mounts	`05_nav_cam_bracket__x1.stl`	1	16 x 40 x 40	6	front webcam

8. Assembly

1. **Pontoons.** Leak-test each segment under water for 60 s. Glue the segments end to end with epoxy and the dowels (bow + mid + stern, per side). Brush a thin epoxy coat over the outside.
2. **Drive motors.** Put each JGA25-370 in its pocket with the gearbox face against the outer wall. Fit 2 × M3 screws through the slots, and put a saddle on top. Then fit the paddle wheels: the hub faces the hull, and an M3 grub screw goes onto the shaft flat.
3. **Deck.** Screw the deck onto the pontoon tops with M3 × 16 self-tapping screws (36 holes). The deck clamps the motor saddles down. Pass the motor wires up through the 10 mm holes.
4. **Belts.** Join links with 84 mm pieces of 1.75 mm filament. Push each piece through, then melt both ends flat with a soldering iron.
    - C1: 29 links, with a cleat link at every 4th position.
    - C2: 39 plain links.
    - Wrap each belt around its drums before closing the last joint. The cleats and the plate's flat face point outward.
5. **Intake (C1).**
    1. Bolt the two frame halves together with the splice plates (M3 × 12, 8 per side). Fit the drums on M6 rod axles, with the drive drum's D-bore on the port side.
    2. Mount the 30 rpm motor on the port wall at the top drum and tighten its grub screw.
    3. Screw the lower stand-offs into the bow pontoons' inner walls (M3 × 16). Screw the legs to the upper walls and to the deck.
6. **Sorting conveyor (C2).** Screw the front frame and the head stands to the deck. Bolt the bed beam's 3 ears through the sorter-frame inner walls. Fit the drums and belt. Mount the 30 rpm motor on the port head stand.
7. **Sorter.**
    1. Screw both sorter frames to the deck.
    2. Drop each flap's bottom pin into its pivot hole, then lay the cover over the flap posts.
    3. Screw the servo bridges onto the cover posts and hang the SG90 servos shaft-down. Put a single-arm horn in each post's slot (M2).
    4. Set each servo to 90° before fitting the horn, with the door closed.
    5. Slide the buckets in from the outside.
8. **Electronics.** Fit the electronics box (port bow deck) and the battery box (starboard bow deck). Mount the Pi camera on the mast over the camera zone, with the LED strip in the slots. Mount the FC-51 behind the window in the starboard wall of the camera zone, and the webcam on the bracket on the port lid. Seal every wire hole with hot glue.

9. Wiring

From	To
3S1P pack → BMS → 10 A fuse → rocker switch	12 V bus
12 V bus	L298N #1 VS, L298N #2 VS (keep their 5 V jumpers), both LM2596 inputs
LM2596 #1 out (5.1 V)	Pi Zero 5 V (pin 2) + GND (pin 6)
LM2596 #2 out (5.5 V)	PCA9685 V+ (servo power) + LED strip
PCA9685 VCC / SDA / SCL	Pi 3.3 V (pin 1) / GPIO2 (pin 3) / GPIO3 (pin 5)
PCA9685 ch 0, 1, 2, 3	servos for bins 1, 2, 3, 4
L298N #1 IN1, IN2, ENA	GPIO5, GPIO6, GPIO12 (remove the ENA jumper). OUT1/2 → both port wheel motors in parallel
L298N #1 IN3, IN4, ENB	GPIO20, GPIO21, GPIO13. OUT3/4 → both starboard wheel motors in parallel
L298N #2 IN1, IN2, ENA	GPIO23, GPIO24, GPIO18. OUT1/2 → intake (C1) motor
L298N #2 IN3, IN4, ENB	GPIO16, GPIO26, GPIO19. OUT3/4 → sorting (C2) motor
FC-51 VCC / GND / OUT	Pi 3.3 V (pin 17) / GND / GPIO17
Pi camera / webcam	Pi Zero camera port (22-pin cable) / the micro-USB data port via the OTG adapter

Connect all grounds together. If a motor runs backward, swap its two wires.

10. Software and camera training

1. Flash Raspberry Pi OS (64-bit) on the Pi Zero 2 W and copy this folder to `~/trash-sorter-boat`. Then run `cd ~/trash-sorter-boat/software && ./setup.sh`. The script installs the libraries, turns on I²C, creates the **TRASHBOAT** Wi-Fi hotspot (password `boat12345`) and starts the app on every boot.
2. **Train the camera.**
    1. Run `python3 collect.py "Plastic bottle"` (then the same for `Can`, `Paper`, `Other` and `Empty`, the bare belt). Take 80 or more photos of each class on the belt.
    2. Upload the photos to Google Teachable Machine (Image project) and train.
    3. Export → TensorFlow Lite → Floating point. Copy `model_unquant.tflite` and `labels.txt` into `software/model/`.
3. **Drive.** Join TRASHBOAT on the phone and open `http://10.42.0.1:8080`. Press **Sorting ON** to run both conveyors and the sorter.
4. **Tune** in `config.py`:
    - door angles: `SERVO_CLOSED`, `SERVO_OPEN`
    - belt run time per bin: `TRAVEL_S`
    - class → bin mapping: `LABEL_TO_BIN`
    - confidence threshold: `MIN_CONFIDENCE`

`python3 test_sorter.py` checks the sorting logic without a Pi. `python3 app.py --mock` runs the dashboard on a laptop.

11. Test plan and assessment

Test	Method	Pass criterion
T1 Leak	Hold each pontoon segment under water for 60 s, then weigh it. Float the full boat for 10 min	No bubble stream; mass gain < 5 g; boxes dry
T2 Draft and stability	Float the boat with 500 g of trash and measure the draft. Press one side down 20°	Draft within ±5 mm of the computed value; self-rights
T3 Collection	Release 20 floating items in front of the bow while driving slowly	≥ 16 of 20 reach the sorting belt
T4 Sorting accuracy	Run 10 items of each class through the sorter	≥ 80% end up in the correct bin
T5 Water and land speed	Time a 3 m run in water, then on concrete	≥ 0.2 m/s in water; ≥ 0.4 m/s on land
T6 Land–water transition	Drive out of the water up a 15° ramp and back in	Completes both ways without help
T7 Failsafe and endurance	Turn the phone's Wi-Fi off while driving; then run until the battery is flat	Drive stops within 1 s; run time ≥ 45 min

Scoring rubric (each criterion 1–4):

Criterion	Weight
Buoyancy and stability	15%
Trash collection rate	20%
Sorting accuracy	20%
Mobility on land and water	15%
Cost efficiency	10%
Documentation	20%

12. Safety

- Use genuine 18650 cells with the BMS and the fuse. Charge only out of the boat, with the 12.6 V charger.
- Keep hands off the belts and wheels while powered. The dashboard's Sorting OFF button stops both conveyors, and the drive stops by itself if the phone link drops.
- On land, turn Sorting OFF: the intake cleats pass only 7 mm above flat ground.
- Use it in calm, supervised water. PLA and PETG don't biodegrade, so recover any broken part.

13. Verification and limitations

Checked in software:

- Interference check of all 66 assembly parts: 0 overlaps, with a door open and with all doors closed.
- All 35 STL files are watertight meshes and fit a 220 × 220 × 250 mm bed.

- The sorting logic passes its unit tests, and the phone dashboard runs in mock mode.

Limits:

- The boat has not been printed or tested. Masses, speeds and run time are estimates.
- The motor face-hole spacing differs by brand, so measure your motors first.
- The official OCNR guidelines were not available. The design follows the brief given by the team.
- Prices are approximate Philippine online prices (October 2026). Check them with the suppliers.

References

1. Halliday, Resnick & Walker. *Fundamentals of Physics* (Archimedes' principle).
2. Tupper, E. C. *Introduction to Naval Architecture* (metacentric height).
3. Google Teachable Machine, teachablemachine.withgoogle.com (image classification, TensorFlow Lite export).
4. Raspberry Pi Ltd. Raspberry Pi Zero 2 W and Camera Module documentation.
5. MakerLab Electronics, makerlab.ph; Circuitrocks, circuit.rocks (component sources).